SPARKY'S SCIENCE EXPERIMENTS

Bottle Rain Gauge: Measure the Rain

Natural Science (weather) · Ages 7+ · 20 minutes, then check every day

Make a rain gauge from a plastic bottle and measure how many millimetres of rain fall where you live.

Safety first

Sparky always says: Never go outside to check your rain gauge in a thunderstorm - lightning is dangerous!

- An adult should cut the bottle.
- Empty it every day so mosquitoes can't breed in it.

You need

- A clear 2-litre plastic bottle with straight sides
- Scissors (adult)
- A ruler and a permanent marker
- A handful of small stones
- Water

Steps

1. An adult cuts off the top of the bottle where the sides become straight.
2. Put the stones in the bottom, then add water until the bumpy bottom is just covered. Mark that level 0.
3. With the ruler, mark lines every 5 mm up the side and number them.
4. Put the cut-off top in upside down, as a funnel.
5. Stand it in an open place, away from trees and roofs.
6. Every morning, read the level, write it down, and pour out water to the 0 line.

If it doesn't work, try these in order

1. Too much rain? Make sure you poured back to the 0 line each day.
2. Too little? Trees or roofs may block the rain - move it into the open.
3. Falls over? Add more stones or push it into the ground.

Why it works

Rain is measured in millimetres: how deep it would be if it stayed where it fell. 1 mm of rain means 1 litre of water on every square metre of ground. The funnel is as wide as the bottle, so the water height in the bottle is the rain that fell, and it stops water evaporating.

Did you know? Most of South Africa gets its rain in summer, often from afternoon thunderstorms, but the Western Cape gets most of its rain in winter. South Africa is dry: about 450 mm a year.

Ask Sparky: How much rain fell this week? Is it rainy season where you live?